Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building and debugging **data-driven applications**, **object-oriented services**, and testing workflows, with strong grounding in C++/Java/Python, systems fundamentals, and technical documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized multi-tenant application workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through structured data design, query tuning, and systematic debugging of backend execution paths.
- Built automated validation and ETL checks for 5,000+ monthly jobs, improving reliability and accelerating issue resolution through earlier detection of regressions, runtime anomalies, and data-quality drift.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested full-stack application features handling 10,000+ daily requests, translating product needs into maintainable backend services and interfaces while improving debugging, release confidence, and documentation quality.
- Implemented secure batch-processing workflows for PDF, DOCX, and ZIP ingestion with AWS-backed persistence, reducing feedback turnaround by **60%** while supporting stable production operations and issue triage.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based experimentation pipelines across 100,000+ records, creating reproducible evaluations and technical artifacts that improved debugging quality, test repeatability, and communication of system behavior.
- Produced technical reports and targeted validation scenarios translating findings into actionable improvements, supporting stronger design reviews, clearer handoffs, and more reliable implementation decisions across collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform API and ETL software with microservice architecture, automated VM testing, and 20+ unit and integration suites using Jenkins, PyTest, and Mockito in collaborative Agile delivery cycles.
- Troubleshot and optimized API integrations using object-oriented design patterns, AWS Glue, and S3, improving response times by **30%** and increasing pipeline efficiency by **68%** across production workloads.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based application with WebSockets and REST APIs, supporting 2M+ synchronized records and six core workflows through modular components, repeatable testing, and maintainable service boundaries.
- Deployed Azure SQL with Jenkins-based CI/CD and automated testing, achieving 95%+ successful build validation while improving release consistency and production-readiness across iterative delivery cycles.

Skills

Languages: **C++**, **Java**, **Python**, JavaScript/TypeScript, SQL, Bash
Software Engineering: object-oriented design, data structures, operating systems, debugging, unit/integration testing, technical documentation
Application Development: REST APIs, Flask, FastAPI, Node.js, React, SQL-backed systems, system design
Tools: Git/GitHub, Docker, Jenkins, AWS, Azure, CI/CD, Linux, collaborative code reviews

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with **C++** logic, Gantt visualization, and dynamic metrics, improving debugging efficiency by **20%** across simulated workloads.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, maintaining Java and TypeScript modules that accelerated schema exploration, query authoring, and structured debugging of data workflows.

HMIS Code | *MERN, TypeScript, ABDM APIs, Sentry*
- Built a microservice-based healthcare management platform supporting 100+ daily appointment workflows, reducing patient-history retrieval by 2x through structured schemas, monitoring, and maintainable service-level design.